

HTX xData Technical Test — Task 4

1. Results

I compared the base NVIDIA Parakeet-TDT-0.6B-v3 model against the fine-tuned models on **The Daily Ketchup (TDK)** clips.

All TDK clips were excluded from training and validation. The fine-tuned model used for the final comparison, **E3**, was selected based on non-TDK validation performance.

Metric	Base	E1	E3 (Selected)
WER	24.4272%	21.3120%	19.0689%
CER	18.1281%	15.8369%	14.2119%
Evaluated clips	218,011	218,011	218,011

E3 reduced WER from **24.43% to 19.07%**, an absolute improvement of **5.36 percentage points** and a relative WER reduction of **21.94%**.

CER also decreased from **18.13% to 14.21%**.

All 218,011 TDK clips across 367 videos were retained in the evaluation. I did not remove high-error clips when calculating these results.

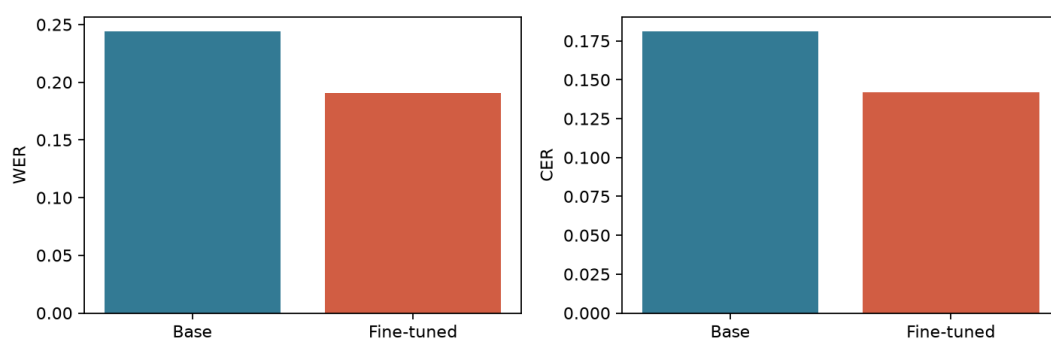


Figure 1. Corpus WER and CER for the base and selected fine-tuned model on the held-out TDK evaluation set.

Both WER and CER improve after fine-tuning, showing that the improvement is visible at both the word and character level.

I also ran a 2,000-draw video-cluster bootstrap on the WER difference. The 95% interval for E3-minus-base WER was **[-5.4961, -5.2234] percentage points**. I clustered by video because clips from the same video should not be treated as fully independent observations.

This interval should still be interpreted specifically in the context of the TDK evaluation set, rather than as a claim about all Singapore English speech.

2. Fine-Tuning Setup

TDK was excluded before constructing the training and validation sets.

The E1/E3 dataset contained:

Split	Clips	Duration	Source videos
Training	10,972	10.046 h	852
Validation	2,133	2.023 h	93

Training and validation were video-disjoint. This prevents clips from the same source video appearing in both sets, although it does not guarantee that the same speaker never appears across the two sets.

Audio was decoded, validated and converted to **mono 16 kHz**. I retained the pretrained model's **8,192-token SentencePiece tokenizer** and **128-bin log-mel frontend**.

I did not apply denoising, VAD trimming, speech-quality filtering or tokenizer replacement.

The main fine-tuning configuration was:

- Optimizer: AdamW
- Learning rate: `1e-5`
- Betas: `(0.9, 0.98)`
- Weight decay: `0.001`
- Precision: BF16
- Batch size: 8
- Gradient accumulation: 4
- Gradient clipping: 1.0
- Seed: 2026

- Validation interval: every 50 optimizer updates

I ran the following main experiments:

Experiment	Training setup	Validation WER
E1	10.046 h, 1,200 updates	26.5402%
E2	25.046 h, 1,200 updates	26.6967%
E3	E1 dataset, 3,000 updates	24.6470%

E3 gave the best valid non-TDK validation WER and was therefore selected for the final TDK comparison.

One useful observation is that **E2 should not be interpreted as evidence that adding more training data made the model worse**. E1 and E2 used the same number of optimizer updates despite E2 having substantially more training data. Therefore, each E2 example received less training exposure on average.

E3 instead kept the E1 dataset and increased the training budget from 1,200 to 3,000 updates. Its better validation WER suggests that additional training was useful for this dataset and configuration.

3. Training Behaviour

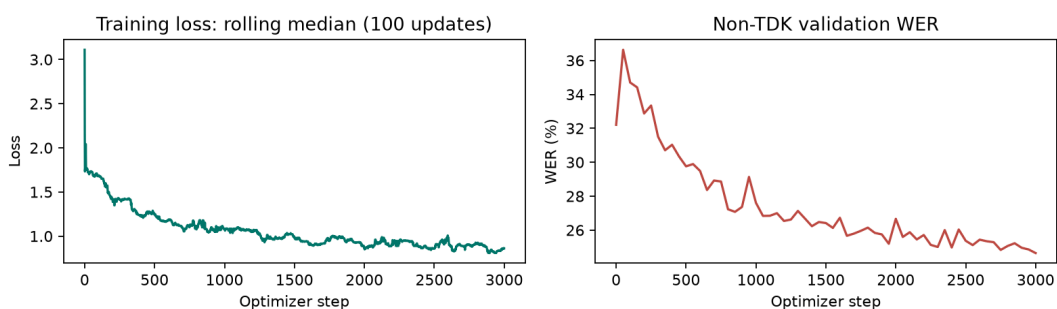


Figure 2. E3 training curves. Left: training loss using a 100-update rolling median. Right: WER on the non-TDK validation set.

Training loss decreases throughout the run. More importantly, non-TDK validation WER also decreases overall, from above 30% during the early validation checks to **24.647%** at the selected E3 checkpoint.

The validation curve fluctuates between checks, but the overall trend is downward.

The best E3 result occurred near the 3,000-update training limit. I therefore cannot conclude from this run alone that training had fully converged. A longer

run with validation-based early stopping would be useful to determine where performance actually plateaus.

E4 continuation issue

I attempted to continue E3 in a subsequent E4 run. However, the resumed run inherited the previous Timer state and stopped after only 20.82 additional fit minutes.

It also produced an inconsistent **14.93% validation WER** at the unchanged resume step, despite the full validation result immediately before continuation being approximately **24.65%**. Later E4 validation checks returned to approximately **24.61-25.28%**.

I therefore excluded E4 from model selection and do not use the 14.93% result as evidence of an improvement.

4. Error Analysis

I grouped the TDK results by clip duration to identify where the model still performs poorly.

Clip duration	Clips	Base WER	E3 WER
<2 s	117,758	49.8087%	37.5281%
2-5 s	62,338	24.2971%	19.2037%
5-10 s	29,134	16.9546%	13.6281%
>=10 s	8,781	13.6598%	10.8692%

E3 improves WER in **every duration group**.

The clearest remaining weakness is very short audio. Clips below two seconds have a base WER of **49.81%** and an E3 WER of **37.53%**. Although fine-tuning provides a large improvement, these clips remain substantially harder than longer utterances.

Possible reasons include incomplete utterances, transcript/audio alignment and overlapping speech. I have not manually labelled these errors, so these should be treated as hypotheses rather than confirmed causes.

At the video level, E3 improved WER over the base model for **all 367 TDK videos**. This suggests that the overall improvement is not being produced by only a small number of particularly successful videos.

5. Proposed Improvements

Based on the results above, I would focus the next experiments on four areas.

5.1 Check data quality and alignment

The `<2 s` group contains the highest WER and also represents a large proportion of the dataset.

I would manually inspect a stratified sample of these clips, together with samples from the other duration groups. The aim would be to determine how much of the remaining error comes from genuinely difficult speech versus issues such as clipped utterances, incorrect timestamps or transcript alignment.

I would not automatically remove examples simply because they have high WER.

5.2 Properly test the effect of additional YCSEP data

E2 used approximately 25 hours of data compared with approximately 10 hours for E1, but both were trained for 1,200 optimizer updates.

This makes it difficult to determine whether the E2 result is caused by the additional data or by reduced training exposure per example.

I would therefore compare the 10-hour and 25-hour datasets under two conditions:

1. **Matched optimizer updates**, and
2. **Matched number of exposures per training example**.

The validation set would remain fixed for both experiments.

This would provide a cleaner answer to whether increasing the amount of YCSEP training data improves the model.

5.3 Tune learning rate and training duration

E3 improved when training was extended from 1,200 to 3,000 updates, and its best validation result occurred near the end of the run.

After fixing the E4 resume/validation issue, I would test a small learning-rate range such as:

`5e-6` , `1e-5` , and `2e-5`

Each run would use the same training and validation manifests and validation-based early stopping.

If compute permits, I would repeat the strongest configurations with multiple random seeds to check that the improvement is reproducible.

5.4 Investigate short utterances

Because clips below two seconds remain the weakest duration group, I would specifically test whether additional context helps them.

One experiment would concatenate adjacent segments when they belong to the same speaker and the transcript alignment can be preserved. The resulting model could then be compared against the unchanged E3 setup.

Audio augmentation or denoising could also be tested, but I would evaluate each change separately rather than combining several preprocessing changes in one experiment. This makes it possible to identify which change actually affects WER.

6. Conclusion

Fine-tuning Parakeet-TDT-0.6B-v3 on non-TDK YCSEP data improved performance on the held-out TDK evaluation set.

The selected E3 model reduced:

- **WER from 24.43% to 19.07%**
- **CER from 18.13% to 14.21%**

This corresponds to a **21.94% relative reduction in WER**.

The improvement was also present across every clip-duration group and all 367 evaluated TDK videos.

The experiments also identify several areas that remain unresolved. In particular, E2 does not provide a controlled comparison of training-data size, E3 reached its best validation result near the training limit, and clips shorter than two seconds remain considerably more difficult than longer clips.

I would therefore prioritize controlled experiments on **training-data exposure, training duration and learning rate, data/alignment quality, and short-utterance handling** before making larger changes to the training pipeline.

References

1. *YouTube Corpus of Singapore English Podcasts (YCSEP)*, Harvard Dataverse. DOI: [10.7910/DVN/B7JRID](https://doi.org/10.7910/DVN/B7JRID).
2. NVIDIA, [nvidia/parakeet-tdt-0.6b-v3](https://huggingface.co/nvidia/parakeet-tdt-0.6b-v3), Hugging Face.